

Probit Offset

standard, 5-member ensemble, evaluated on 2016-2026, measured against the market

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Model identity

Every figure below is read from a declared artifact. This report fits nothing, loads no booster, recomputes no prediction and calls no betting optimiser.

MODEL Probit Offset standard	EVALUATION YEARS 11 2016-2026	ENSEMBLE MEMBERS 5 seeds 7, 17, 42, 123, 256	BOOSTING ROUNDS 100
FEATURE SET production 28 declared inputs			

Probit Offset is not the production model.

It is one of the standard published architectures, measured against the market on the same races as every other report here. Nothing in it is a promotion decision, and the promotion gates it can and cannot satisfy are in the technical appendix.

HOW EACH EVALUATION YEAR WAS FITTED

These are walk-forward figures rather than one model's: the model is refitted for each evaluation year on the seasons before it, so the result concatenates 11 separately fitted models and no evaluation year's outcomes appear in its own fit. This run did not record the year its training window starts, so this report does not name one.

Deltas in this report use `model_minus_market_log_loss: race_multinomial_log_loss`, where **negative** is better. That is stated because three delta fields existed in this repository and they did not all point the same way.

Provenance -- the source commit, the config and data digests, the environment and every input this report read by content digest -- is in the technical appendix and in `run_lock.json` beside this file. Canonical publication fails without it.

Executive summary

MEAN DELTA VS MARKET -0.004358 negative is better	RACES 7,933	YEARS BETTER 10/11	PAIRED DELTA -0.000943 against softmax_offset
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Probit Offset is better than the market by 0.004358 `race_multinomial_log_loss` on 7,933 races. The direction holds in 10 of 11 evaluated years. There is also a paired diagnostic against `softmax_offset`, which holds everything but one change fixed: on that comparison the delta is -0.000943. Every seed agrees on the sign of that paired delta.

Evaluation population and market reference

THE MARKET REFERENCE

Every metric below is reported beside the market's own implied probabilities on the same races, normalised within each race, and the delta is this model's race log loss minus the market's on the same races. One yardstick across every published report, so their headline numbers mean the same thing and can be read against one another. The market is a measurement, not a model this one is defined against.

THE EVALUATION POPULATION

Every figure in this report covers the standardized evaluation window, which begins in 2016. One window for every published report, so their headline numbers are computed on the identical race keys and can be compared -- which is checked by digest before anything is published, not asserted in a footnote. 2015 is earlier than the window and is excluded from every evaluation figure below, including the bankroll paths. The feature-effect grid is the one exception, and says so where it appears: a learned response direction is a property of each year's fit rather than of the evaluation population.

TEST RACES
7,933

YEARS
11
2016-2026

SEEDS
5

Some years are warm-up rather than evaluation

2015 had no earlier year to fit a utility scale on and took the predeclared theoretical value (insufficient_history). The utility is divided by that scale, so those years amplify corrections where a fitted scale halves them, and their stakes and claimed edges are several times any fitted year's. They are excluded from every figure in this report and named here rather than labelled and left in.

Overall probability performance

races	model log loss	market log loss	market delta	better year share
7933	2.024447	2.028805	-0.004358	0.909091

One row, pooled over every evaluated race, and measured against the market: market_delta is the model's race log loss minus the market's on the same races, so negative means the model is better. The per-race distribution is what the year-by-year section shows; a pooled mean can be carried by a handful of races and this figure alone cannot tell you whether it was.

Year-by-year probability performance

test year	races	model log loss	market log loss	market delta	honest
2016	744	2.092697	2.099277	-0.006580	yes
2017	772	2.017573	2.021559	-0.003986	yes
2018	756	2.056572	2.061504	-0.004932	yes
2019	759	1.926170	1.932827	-0.006656	yes

test year	races	model log loss	market log loss	market delta	honest
2020	799	2.006427	2.010400	-0.003974	yes
2021	783	2.001404	2.006659	-0.005255	yes
2022	783	2.050094	2.051195	-0.001100	yes
2023	796	2.020445	2.024066	-0.003621	yes
2024	792	2.017508	2.025744	-0.008236	yes
2025	748	2.051190	2.057397	-0.006207	yes
2026	201	2.028838	2.026224	0.002614	yes



Ahead of the market in 10 of 11 years.

Calibration and sharpness

This architecture reports calibration through a paired comparison rather than binned reliability: this model and its reference are each given a temperature on the same races, so the calibrated delta is the like-for-like figure. The reference is named in the paired diagnostic in the technical appendix, together with what the comparison holds fixed. Fallback-scale years are excluded here, as they are from the headline.

test year	this model calibrated loss	reference calibrated loss	calibrated delta
2016	2.092581	2.093236	-0.000655
2017	2.018646	2.018225	0.000421

test year	this model calibrated loss	reference calibrated loss	calibrated delta
2018	2.057226	2.057516	-0.000290
2019	1.928633	1.927971	0.000662
2020	2.007505	2.009214	-0.001709
2021	2.000953	2.001587	-0.000633
2022	2.050874	2.051204	-0.000330
2023	2.021237	2.021954	-0.000718
2024	2.017259	2.017686	-0.000427
2025	2.053116	2.053894	-0.000779
2026	2.029934	2.030893	-0.000959

If the calibrated delta is materially smaller than the uncalibrated one, part of this model's advantage was a temperature the reference could also have had.

Top-choice movement

Not applicable to this model. Top-choice movement needs the market's own first pick beside the model's, and this run's artifacts do not carry it. The paired diagnostic in the appendix holds the mean utilities fixed, which is a different and narrower question.

Ensemble stability

Not applicable to this model. There is no ensemble to be stable. This architecture is one choice link applied to one set of mean utilities, not a bag of independently fitted members, so seed variation here is integration noise rather than model disagreement -- and that is reported in the appendix instead.

Feature and correction effects

Not applicable to this model. There are no per-feature effects belonging to this model to attribute. A choice link changes how utilities become probabilities; it has no features of its own. The mean utilities are another run's fit, so attributing their feature effects here would credit this model with that run's work -- and for the market-anchored architecture they are the Softmax Offset margins exactly, whose per-feature effects, response directions and group ablation are in that report. What is specific to this model is the link and its utility scale, and both are in the technical appendix.

Betting and bankroll analysis

Retrospective odds

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Do not select a threshold from these returns.

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

Probit Offset is not the production model, and this is not a proposal to stake it.

A settled ledger measures what a probability was worth economically, which is a different question from which model should carry money. That decision is the production model's, and the comparison that would inform it is in the technical appendix rather than in this table.

SETTLEMENT BY STAKING RULE

Role	What it is for
Primary	what an operator would run: sized for the rebate, and credited it.
Sizing reference	the same settlement, sized blind. It isolates sizing for the cliff from receiving the rebate -- this row and the primary settle identically and differ only in what the solver knew.
Settlement ablation	blind sizing with the rebate not credited. It isolates the rebate's own contribution to the return.
Benchmark	a fixed flat stake above an edge floor, sized from the opening bankroll and never resized. Not a Kelly fraction, so it belongs to no preset panel below -- and it has no ruin stop, so a model that finds edge in most races can stake more than the bankroll holds and finish below zero. That is the yardstick behaving as defined rather than a settlement error: it is here to show what indiscriminate staking on this model's edges would have done, which is why it is a benchmark and not a proposal.

The table below is ordered by that role rather than alphabetically, so the first Kelly rows are the ones an operator would run.

strategy	n bets	turnover	net return	roi	total log growth	max drawdown	final bankroll	underwater days
rebate_aware_kelly@0.01	3646	8,722,790	338914.500000	0.038854	0.079050	0.038756	10822587.500000	2206
rebate_aware_kelly@0.02	3797	21,312,520	989793.500000	0.046442	0.212640	0.080152	12369387.900000	2298
rebate_aware_kelly@0.05	5038	74,758,690	1254079.500000	0.016775	0.511552	0.187908	16678771.850000	3262
rebate_aware_kelly@0.1	6573	202,278,270	1252534.500000	0.006192	0.974015	0.341975	26485578.700000	4692
kelly_settled_rebate@0.01	3319	5,587,210	448962.000000	0.080355	0.058282	0.030273	10,600,136	2211
kelly_settled_rebate@0.02	3343	11,632,560	904,457	0.077752	0.138195	0.054747	11,481,996	2228
kelly_settled_rebate@0.05	3351	32,396,920	2255565.500000	0.069623	0.351941	0.127529	14218249.500000	2234
kelly_settled_rebate@0.1	3355	75,650,170	4259989.500000	0.056312	0.657041	0.239877	19290763.500000	2238
kelly_no_rebate@0.01	3318	5,540,860	446268.500000	0.080541	0.043660	0.032880	10446268.500000	2210
kelly_no_rebate@0.02	3343	11,272,550	869024.000000	0.077092	0.083332	0.064842	10,869,024	2232
kelly_no_rebate@0.05	3350	29,435,390	2,012,729	0.068378	0.183382	0.155179	12,012,729	2239
kelly_no_rebate@0.1	3352	61,666,500	3353997.000000	0.054389	0.289231	0.288621	13,353,997	2240
flat_stake@edge>=0.02	2655	26,550,000	2,592,000	0.097627	0.230477	0.132261	12,592,000	1746

Model view bagged_probit_offset. Seed -1. These rules see the same races and the same odds. They do not necessarily choose the same runners or the same races: rebate-aware sizing admits runners a blind solve would reject, because staking a mildly negative-edge runner raises the loss on every other outcome and can push a race over the rebate threshold. Compare the bet counts as well as the returns.

What this settlement can decide

Secondary diagnostic. This suite selects on paired race log loss; betting returns cannot select a model and are not evidence for or against a probit candidate.

EXPECTED AGAINST REALISED, BY ESTIMATED EDGE

FLAT STAKE

bucket	mean estimated edge	expected return	realised at edge>=0.02	bets at edge>=0.02
0.020000 to 0.050000	0.033997	0.033997	-0.0434	718
0.050000 to 0.100000	0.072664	0.072664	0.0151	610
0.100000 to 0.200000	0.143784	0.143784	0.0230	561
0.200000 to unbounded	0.511528	0.511528	0.3502	766

KELLY, NO REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.009690	0.009690	-0.1013	-0.0979	-0.0877	-0.0766	663	688	695	697
0.020000 to 0.050000	0.033997	0.033997	0.0176	0.0170	0.0185	0.0206	718	718	718	718
0.050000 to 0.100000	0.072664	0.072664	0.0542	0.0536	0.0528	0.0501	610	610	610	610
0.100000 to 0.200000	0.143784	0.143784	-0.1521	-0.1528	-0.1546	-0.1573	561	561	561	561
0.200000 to unbounded	0.511528	0.511528	0.2149	0.2092	0.1937	0.1688	766	766	766	766

KELLY, SETTLED REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.009677	0.009677	-0.1025	-0.0939	-0.0788	-0.0586	664	688	696	700
0.020000 to 0.050000	0.033997	0.033997	0.0156	0.0163	0.0188	0.0217	718	718	718	718
0.050000 to 0.100000	0.072664	0.072664	0.0549	0.0547	0.0556	0.0546	610	610	610	610
0.100000 to 0.200000	0.143784	0.143784	-0.1523	-0.1534	-0.1562	-0.1599	561	561	561	561
0.200000 to unbounded	0.511528	0.511528	0.2150	0.2107	0.1969	0.1738	766	766	766	766

REBATE-AWARE KELLY

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
unbounded to 0	-0.032218	-0.032218	-0.0699	-0.0459	-0.0344	-0.0230	327	453	1688	3227
0 to 0.020000	0.009677	0.009677	-0.1497	0.0506	-0.0037	-0.0092	664	689	695	691
0.020000 to 0.050000	0.033997	0.033997	0.0553	0.0334	0.0225	0.0021	718	718	718	718
0.050000 to 0.100000	0.072664	0.072664	0.0147	0.0384	0.0131	-0.0108	610	610	610	610
0.100000 to 0.200000	0.143784	0.143784	-0.1503	-0.1433	-0.1623	-0.1283	561	561	561	561
0.200000 to unbounded	0.511528	0.511528	0.1804	0.1787	0.1646	0.1380	766	766	766	766

The check that matters most in this section: if the realised return in each edge bucket does not track the expected one, the edge estimate is wrong and every stake sized from it is wrong with it. The bucket counts are small enough that one winner moves a bucket, so read the pattern across buckets rather than any single number.

A negative-edge bucket appears only under rebate-aware sizing, and only at the larger fractions. That is the rule working as defined rather than a filter failing: the rebate is a cliff, so a runner whose own edge is negative can pay for itself by pushing the race's turnover over the threshold, and the stakes only get large enough for that at the higher fractions.

KELLY PRESETS

OPENING BANKROLL HK\$10,000,000	PRESETS SETTLED 1% Kelly, 2% Kelly, 5% Kelly, 10% Kelly	PATH MODES annual reset, continuous	SEEDS 5 7, 17, 42, 123, 256
TRAINED ON 11 separate fits, one per evaluation year the training start year is not recorded in this run's metadata	BOOSTING ROUNDS 100	YEARS STAKED 2016-2026 2015 excluded	

These are walk-forward figures rather than one model's: the model is refitted for each evaluation year on the seasons before it, so the result concatenates 11 separately fitted models and no evaluation year's outcomes appear in its own fit. This run did not record the year its training window starts, so this report does not name one. Each path is rebased on its own opening balance, so zero means break-even for every line from the same point. These curves are drawn from a bounded 998-point subsequence of up to 4,026 staked races, keeping each interval's highest and lowest balance so no peak or drawdown is smoothed away, and keeping the first and last points exactly. Every figure in the tables is computed from the full settlement, and the complete per-race path is in `race_paths.parquet`.

Where a year is excluded, a continuous path opens at the balance the bank held when the kept years begin and an annual-reset path opens at the declared bankroll, so the two show different opening figures for the same rule. Rebate policy: 10% of the race net loss, once that loss reaches HK\$10,000, as a cliff. Flat staking is omitted from these panels. Its stake is a fixed amount per bet and this settlement never halts on ruin, so its path can pass below zero -- a measure of how badly a rule loses rather than a balance an account could hold. It is also not a Kelly fraction, so it belongs to no preset. Its settlement is in the strategy table above. 2015 is excluded from every path below. These test years have no earlier year to fit the utility scale on and fall back to the predeclared theoretical value. The utility is divided by the scale, so the fallback amplifies corrections where a fitted scale halves them: their stakes and claimed edges are several times any fitted year's and are not comparable with them. Exclude these years from any return figure -- filter on is_honest_scale in the per-run CSV.

1% KELLY

OPENING BANKROLL

HK\$10,859,819

PRIMARY FINAL RETURN

-0.343%

Rebate-aware Kelly, continuous

BETS

1,682

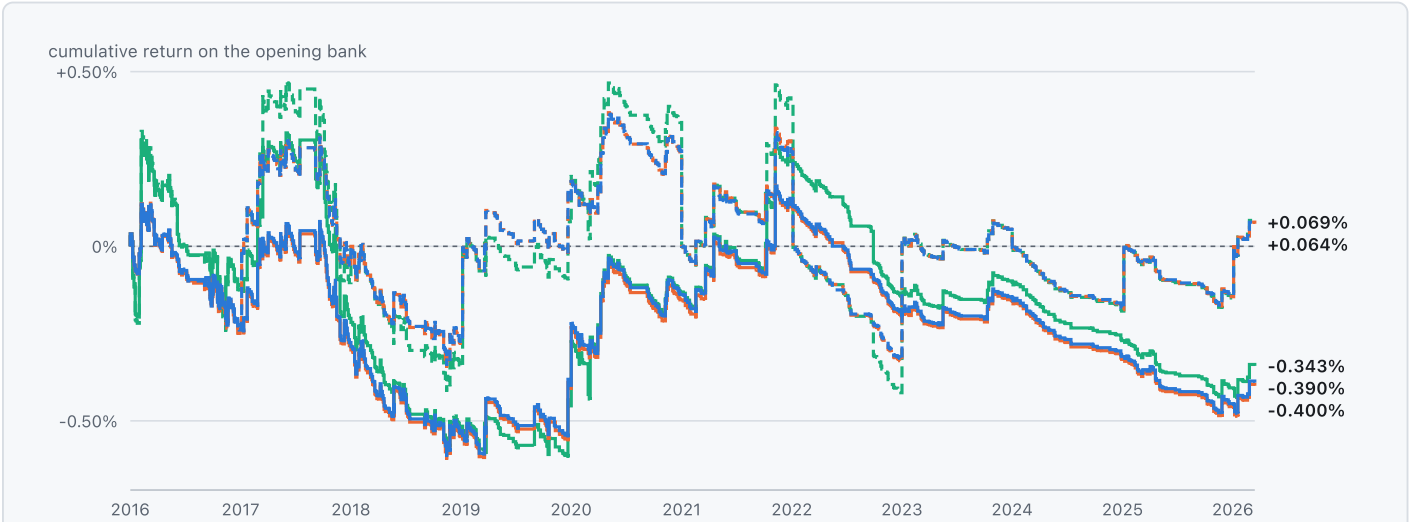
settled bets

REBATE RECEIVED

HK\$14,000

by the primary rule

Rebate-aware Kelly Kelly, settled rebate Kelly, no rebate



Cumulative bankroll return at 1% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	998	+0.069%	1,682 bets
Rebate-aware Kelly, continuous	998	-0.343%	1,682 bets
Kelly, settled rebate, annual reset	998	+0.064%	1,672 bets
Kelly, settled rebate, continuous	998	-0.400%	1,672 bets
Kelly, no rebate, annual reset	998	+0.064%	1,671 bets
Kelly, no rebate, continuous	998	-0.390%	1,671 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	254	253	253
2017	286	284	284
2018	233	231	231
2019	161	160	160
2020	196	193	193
2021	150	150	149
2022	142	141	141
2023	79	79	79
2024	81	81	81
2025	80	80	80
2026	20	20	20

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 254 bets in 2016 against 80 in 2025. Kelly, settled rebate: 253 bets in 2016 against 80 in 2025. Kelly, no rebate: 253 bets in 2016 against 80 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$14,000; Rebate-aware Kelly, continuous HK\$14,000. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

2% KELLY

OPENING BANKROLL

HK\$11,970,655

PRIMARY FINAL RETURN

+3.3%

Rebate-aware Kelly,
continuous

BETS

1,835

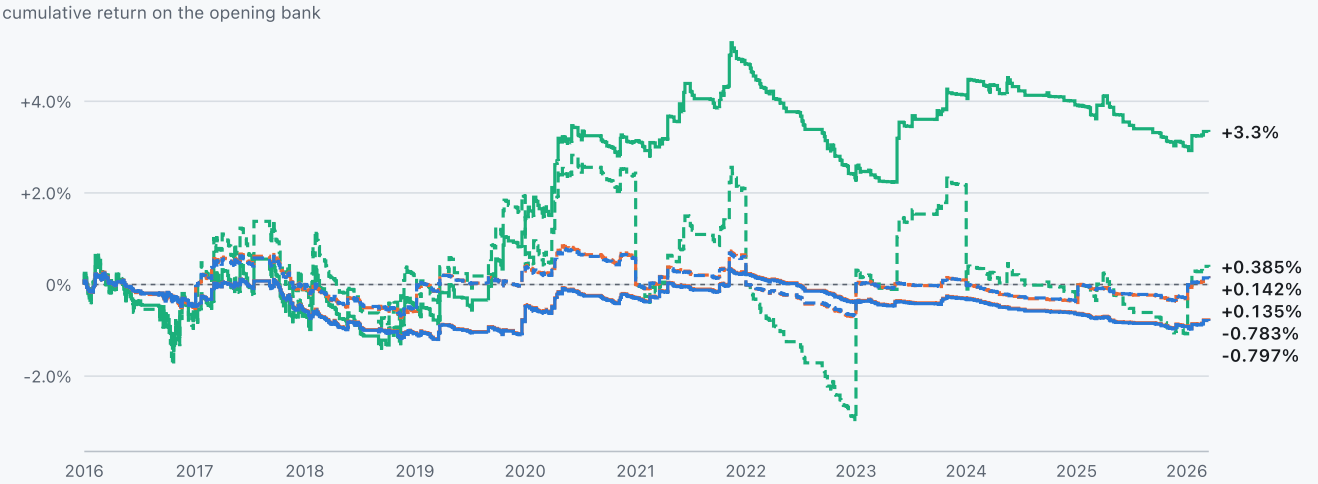
settled bets

REBATE RECEIVED

HK\$315,826

by the primary rule

Rebate-aware Kelly Kelly, settled rebate Kelly, no rebate



Cumulative bankroll return at 2% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	998	+0.385%	1,835 bets
Rebate-aware Kelly, continuous	998	+3.3%	1,835 bets
Kelly, settled rebate, annual reset	998	+0.142%	1,696 bets
Kelly, settled rebate, continuous	998	-0.783%	1,696 bets
Kelly, no rebate, annual reset	998	+0.135%	1,696 bets
Kelly, no rebate, continuous	998	-0.797%	1,696 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	286	254	254
2017	311	287	287
2018	248	233	233
2019	172	162	162
2020	213	198	198
2021	162	153	153
2022	157	144	144
2023	88	82	82
2024	93	83	83
2025	85	80	80
2026	20	20	20

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 286 bets in 2016 against 85 in 2025. Kelly, settled rebate and Kelly, no rebate: 254 bets in 2016 against 80 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$315,826; Rebate-aware Kelly, continuous HK\$315,826; Kelly, settled rebate, annual reset HK\$1,007; Kelly, settled rebate, continuous HK\$1,007. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its

bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

5% KELLY

OPENING BANKROLL

HK\$14,916,934

PRIMARY FINAL RETURN

+11.8%

Rebate-aware Kelly,
continuous

BETS

3,070

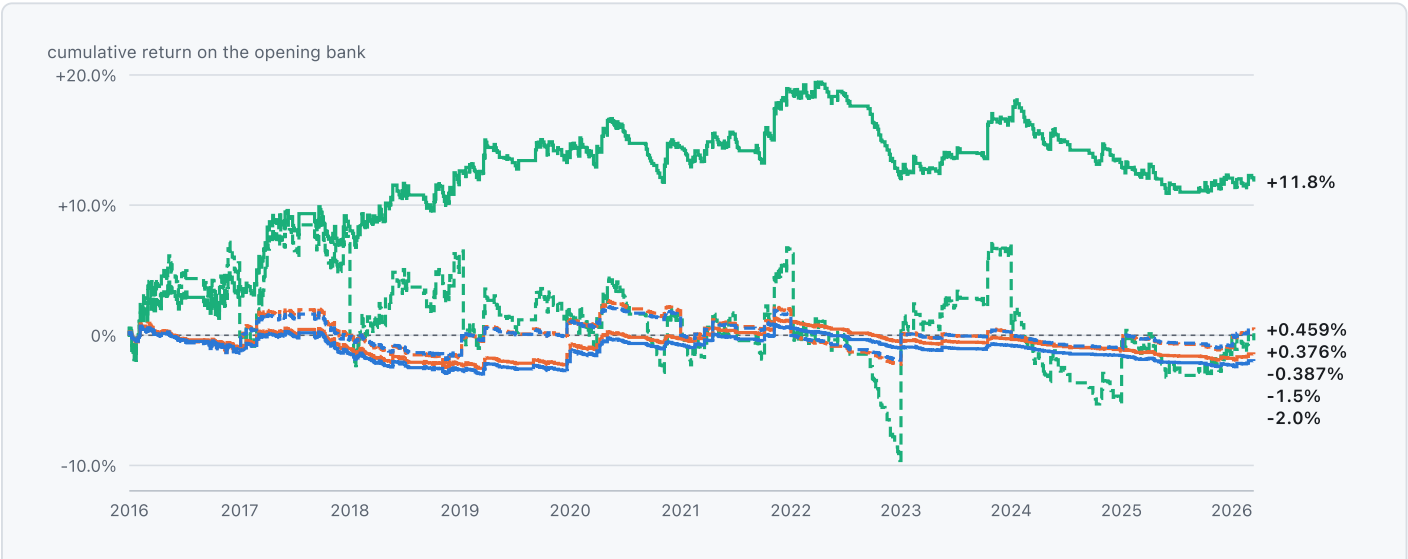
settled bets

REBATE RECEIVED

HK\$2,478,032

by the primary rule

Rebate-aware Kelly Kelly, settled rebate Kelly, no rebate



Cumulative bankroll return at 5% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	998	-0.387%	3,070 bets
Rebate-aware Kelly, continuous	998	+11.8%	3,070 bets
Kelly, settled rebate, annual reset	998	+0.459%	1,704 bets
Kelly, settled rebate, continuous	998	-1.5%	1,704 bets
Kelly, no rebate, annual reset	998	+0.376%	1,703 bets
Kelly, no rebate, continuous	998	-2.0%	1,703 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	411	255	255
2017	460	290	289
2018	373	233	233
2019	266	163	163
2020	335	199	199
2021	295	154	154

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2022	274	145	145
2023	186	82	82
2024	218	83	83
2025	201	80	80
2026	51	20	20

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 411 bets in 2016 against 201 in 2025. Kelly, settled rebate: 255 bets in 2016 against 80 in 2025. Kelly, no rebate: 255 bets in 2016 against 80 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$2,478,032; Rebate-aware Kelly, continuous HK\$2,478,032; Kelly, settled rebate, annual reset HK\$75,876; Kelly, settled rebate, continuous HK\$75,876. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

10% KELLY

OPENING BANKROLL

HK\$20,257,323

PRIMARY FINAL RETURN

+30.7%

Rebate-aware Kelly, continuous

BETS

4,596

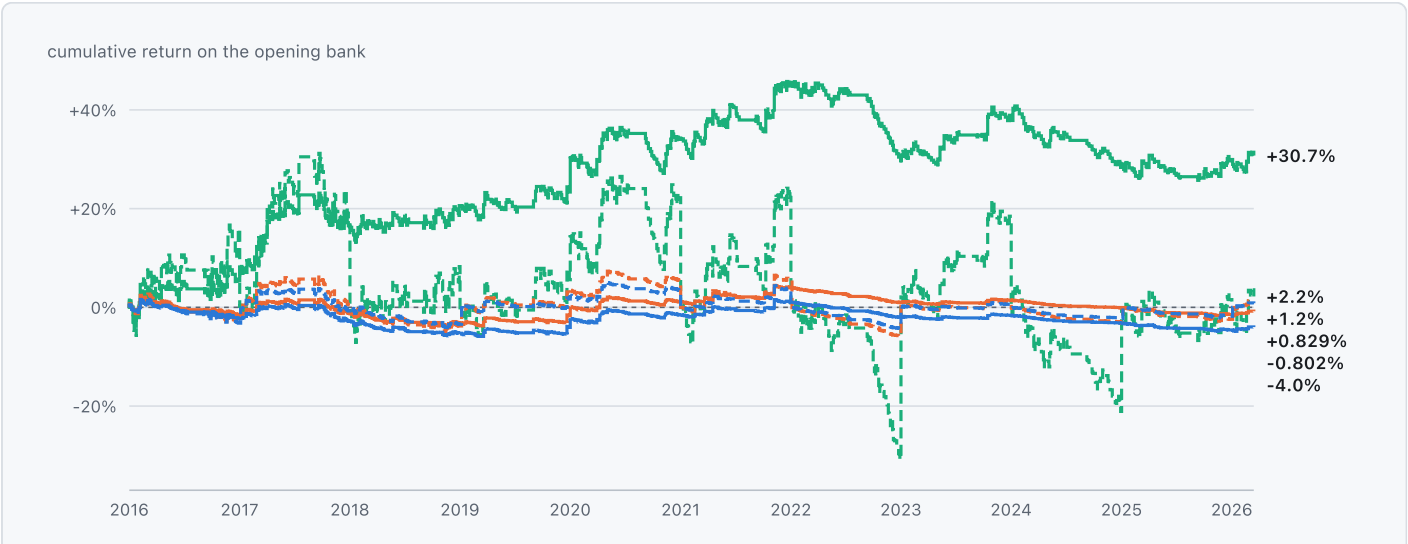
settled bets

REBATE RECEIVED

HK\$8,492,883

by the primary rule

Rebate-aware Kelly
Kelly, settled rebate
Kelly, no rebate



Cumulative bankroll return at 10% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	998	+2.2%	4,596 bets
Rebate-aware Kelly, continuous	998	+30.7%	4,596 bets

Series	Settlements	Final return	Note
Kelly, settled rebate, annual reset	998	+1.2%	1,708 bets
Kelly, settled rebate, continuous	998	-0.802%	1,708 bets
Kelly, no rebate, annual reset	998	+0.829%	1,705 bets
Kelly, no rebate, continuous	998	-4.0%	1,705 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	529	255	255
2017	583	290	290
2018	498	234	234
2019	407	163	163
2020	492	199	199
2021	455	154	154
2022	429	146	145
2023	351	82	82
2024	372	83	83
2025	379	80	80
2026	101	22	20

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 529 bets in 2016 against 379 in 2025. Kelly, settled rebate: 255 bets in 2016 against 80 in 2025. Kelly, no rebate: 255 bets in 2016 against 80 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$8,492,883; Rebate-aware Kelly, continuous HK\$8,492,883; Kelly, settled rebate, annual reset HK\$638,228; Kelly, settled rebate, continuous HK\$638,228. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

PRESET SENSITIVITY

Preset	Rule and mode	Opening	Final	Return	Bets	Rebate
1% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,006,394	+0.064%	1,671	HK\$0
1% Kelly	Kelly, no rebate, continuous	HK\$10,487,212	HK\$10,446,268	-0.390%	1,671	HK\$0
1% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,006,433	+0.064%	1,672	HK\$0
1% Kelly	Kelly, settled rebate, continuous	HK\$10,642,674	HK\$10,600,136	-0.400%	1,672	HK\$0
1% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,006,910	+0.069%	1,682	HK\$14,000
1% Kelly	Rebate-aware Kelly, continuous	HK\$10,859,819	HK\$10,822,587	-0.343%	1,682	HK\$14,000
2% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,013,548	+0.135%	1,696	HK\$0
2% Kelly	Kelly, no rebate, continuous	HK\$10,956,325	HK\$10,869,024	-0.797%	1,696	HK\$0
2% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,014,152	+0.142%	1,696	HK\$1,007
2% Kelly	Kelly, settled rebate, continuous	HK\$11,572,600	HK\$11,481,996	-0.783%	1,696	HK\$1,007
2% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,038,511	+0.385%	1,835	HK\$315,826
2% Kelly	Rebate-aware Kelly, continuous	HK\$11,970,655	HK\$12,369,388	+3.3%	1,835	HK\$315,826
5% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,037,602	+0.376%	1,703	HK\$0
5% Kelly	Kelly, no rebate, continuous	HK\$12,254,926	HK\$12,012,729	-2.0%	1,703	HK\$0

Preset	Rule and mode	Opening	Final	Return	Bets	Rebate
5% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,045,898	+0.459%	1,704	HK\$75,876
5% Kelly	Kelly, settled rebate, continuous	HK\$14,428,220	HK\$14,218,250	-1.5%	1,704	HK\$75,876
5% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$9,961,312	-0.387%	3,070	HK\$2,478,032
5% Kelly	Rebate-aware Kelly, continuous	HK\$14,916,934	HK\$16,678,772	+11.8%	3,070	HK\$2,478,032
10% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,082,865	+0.829%	1,705	HK\$0
10% Kelly	Kelly, no rebate, continuous	HK\$13,911,338	HK\$13,353,997	-4.0%	1,705	HK\$0
10% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,124,254	+1.2%	1,708	HK\$638,228
10% Kelly	Kelly, settled rebate, continuous	HK\$19,446,802	HK\$19,290,764	-0.802%	1,708	HK\$638,228
10% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,218,614	+2.2%	4,596	HK\$8,492,883
10% Kelly	Rebate-aware Kelly, continuous	HK\$20,257,323	HK\$26,485,579	+30.7%	4,596	HK\$8,492,883

The spread across presets is the point: a result that appears at only one Kelly fraction is a result about that fraction rather than about the model. A larger fraction stakes more of the bank on the same edge estimate, so it amplifies both the edge and the error in it.

Simulation and uncertainty

MEAN DELTA

-0.000844

INTERVAL

-0.0012 to
-0.0004

80% level

EXCLUDES ZERO
FAVOURABLY

yes

RACE DAYS

891

39,665 races

REPLICATES

2,000

seed 20260824

A day-block bootstrap: race days are resampled within each test year, so the interval accounts for clustering within a meeting, and the interval lies entirely below zero. It conditions on the fitted model and does *not* include refit uncertainty.

What a simulation cannot tell you

The model-assumed simulation draws winners from the model's own probabilities. If those probabilities are wrong the simulation is confidently wrong in the same direction, so it bounds sampling risk and says nothing about model risk.

THE UTILITY SCALE

FITTED MINIMUM

1.9932

FITTED MAXIMUM

2.2280

THEORY

0.7797

the value theory predicts

The fitted scale sits above the value theory predicts, in every year: 1.993 to 2.228 against 0.780 (the scale the fallback year used, which is the predeclared theoretical value). Theory gets this wrong and the fit is what the report uses -- a scale asserted from theory would divide the utilities by too little, and at that value the market-relative candidate loses instead of winning. A run that took the theoretical value on faith would have reported the wrong sign, which is why the first year -- which has no history and falls back to it -- is excluded from the headline and labelled.

CONFIGURATION

Item	Value
Architecture	probit_offset
Paired-diagnostic reference	softmax_offset
Integration	exact_gauss_hermite_quadrature
Shared factors	0
Opening bankroll	10,000,000

PAIRED ARCHITECTURE DIAGNOSTIC

REFERENCE MODEL

softmax_offset

MODEL LOG LOSS

2.024447

REFERENCE LOG LOSS

2.025390

MODEL MINUS
REFERENCE

-0.000943

negative is better

CALIBRATED

-0.000493

both sides given a
temperature

YEARS BETTER

11/11

WORST YEAR

-0.000049

What this comparison holds fixed: the correction. It is the incumbent's persisted margin, reused exactly rather than refitted, and the market anchor means a zero correction returns the market probability -- so the only difference is the link. A diagnostic, not an identity. This report's headline is the market delta, because that is the one quantity every published report can be compared on. This delta answers a narrower question and answers it better -- it is the only measurement that separates one change from everything else -- and read as a headline it would mislead, because a model can beat the reference it holds fixed and still lose to the market.

PROMOTION GATES

Gate	Verdict
1. paired delta is negative	pass
2. interval excludes zero favourably	pass
3. no year degrades beyond tolerance	pass
4. survives nested calibration	pass
5. direction stable across seeds	pass
6. improvement exceeds integration error 10x	pass

6 of 6 checkable gate assertions hold. Passing them is not promotion. An interaction diagnostic across predeclared cohorts, reproduction from a clean identified commit, and confirmation on a future prospective period are not derivable from these artifacts. Production remains the five-seed Softmax Offset ensemble until they are met.

PER-SEED DELTA

Seed	Delta
7	-0.000947
17	-0.000919
42	-0.000889
123	-0.000943
256	-0.001014

5 of 5 seeds favour the candidate, and every seed agrees on the sign. Shown individually rather than only pooled, because a mean whose seeds disagree is a different claim from one whose seeds agree.

UTILITY SCALE

setting	value	note
2015	0.779697	fitted on strictly earlier years; fallback to the theoretical value
2016	1.993223	fitted on strictly earlier years
2017	2.034700	fitted on strictly earlier years
2018	2.210128	fitted on strictly earlier years
2019	2.226151	fitted on strictly earlier years
2020	2.123985	fitted on strictly earlier years
2021	2.160718	fitted on strictly earlier years
2022	2.145303	fitted on strictly earlier years
2023	2.227982	fitted on strictly earlier years
2024	2.216202	fitted on strictly earlier years
2025	2.093658	fitted on strictly earlier years
2026	2.051702	fitted on strictly earlier years

12 row(s) from the sensitivity artifact, in 1 study/studies. Where a study varies an integration setting, a result that moves with it is a property of the integration rather than of the model.

PROVENANCE

Item	Value
Source commit	eb0e98021dda

Item	Value
Source tree	dirty at generation
Config digest	unrecorded
Identity source	artifact
Model view	bagged_probit_offset
Artifact root	results/probit_models
Inputs read	8 artifact(s), locked by digest in run_lock.json
Evaluation race keys	7933 races, sha256:342d95ee674808a0d7ae4d0df5b0f958da1898863e25c2898bb71d9f58392e58
Environment: numpy	2.5.2
Environment: pandas	3.0.5
Environment: python	3.12.11
Environment: xgboost	3.4.1

The generating source tree was dirty

This run's provenance records uncommitted changes, so the commit named above does not fully describe the code that produced these numbers. Treat the figures as indicative until the report is regenerated from a clean tree.

Every input this report read is listed in `run_lock.json` by content digest, together with the lock digest of the run that produced them -- which is what chains a published figure back through the report to the fit and from there to the feature table's own digest.

Limitations and the retrospective-odds warning

Retrospective odds, restated

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Restated at the end as well as beside the betting numbers, deliberately. It is the single most load-bearing caveat in this report and the one a reader skimming for a return figure is most likely to miss.

Every test year is also development data

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

WHAT ELSE THIS REPORT DOES NOT ESTABLISH

- It measures Probit Offset against the market on the same races. It does not measure this model against the other published models except through that shared yardstick.
- Its intervals condition on the fitted model, so they bound sampling risk and not model risk.
- Every figure is out of sample in time but in sample in *design*: the feature set, the hyperparameters and the staking rules were all chosen with these years visible.
- Nothing here is a shadow test. A profitability claim needs a locked model run forward against timestamped pre-race prices.

Unverified source state

This run recorded a dirty source tree, so its numbers cannot be reproduced from the named commit alone.